

Spatio-Temporal Modeling of Milk Production with Neural Networks

Multi-Resolution Forecasting and What-If Scenarios with a Dynamic Bayesian Network

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Problem

Weekly milk collection forecasting for a dairy cooperative, at the **collection-zone** level.

Predictive challenge

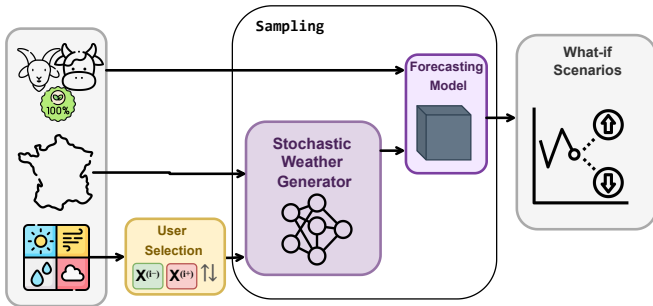
Combining different temporal resolutions — weekly milk and daily weather — without losing short-lived events such as heat waves

Research question

Can a neural forecaster be made queryable by domain experts, despite being a purely correlational, non-causal model?

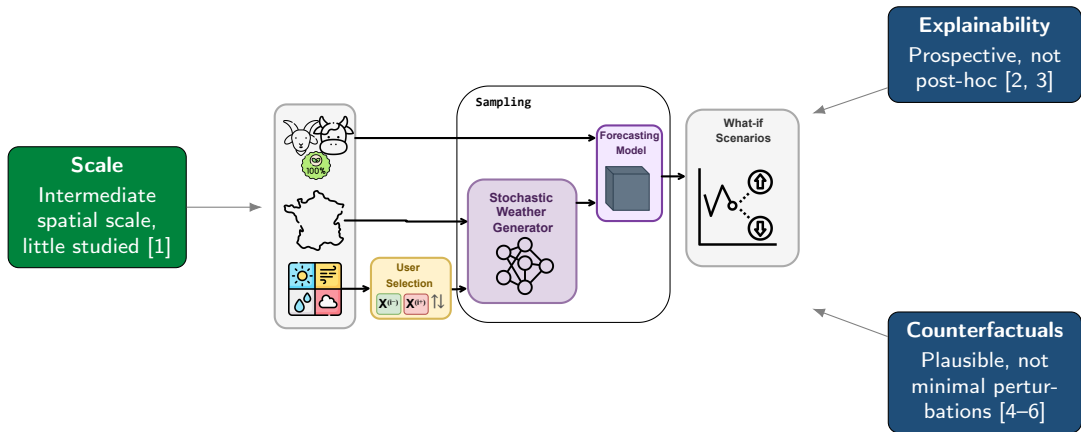
These two challenges are connected: meaningful what-if scenarios require a reliable forecasting model

Pipeline Overview

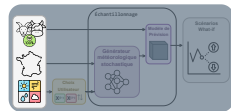


Two contributions : (i) a forecasting model combining milk history and daily weather ; (ii) a stochastic weather generator producing plausible what-if scenarios, propagated through the model

Related Work



Data



Milk delivery

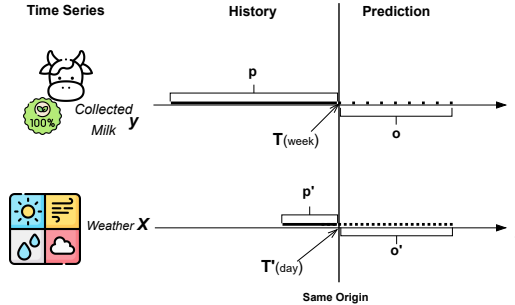
- weekly resolution, 2019–2025
- 17 raw series (zone $\times$ milk type)

Test year : 2025, independently flagged as atypical (heat waves, logistics reorganisation)

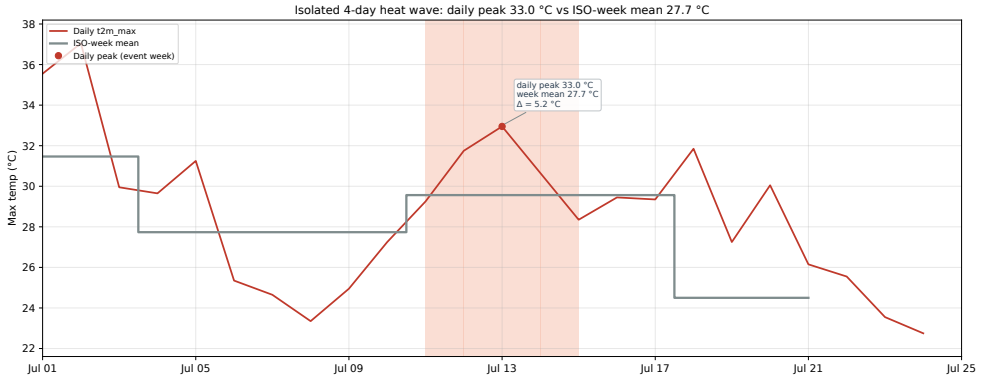
Weather (Weenat)

- daily resolution, 7 variables
- matched to each collection zone

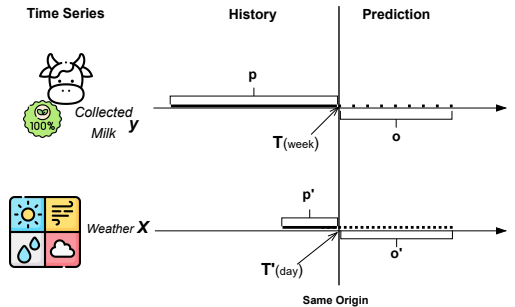
Notation



Notation



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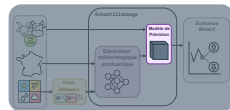
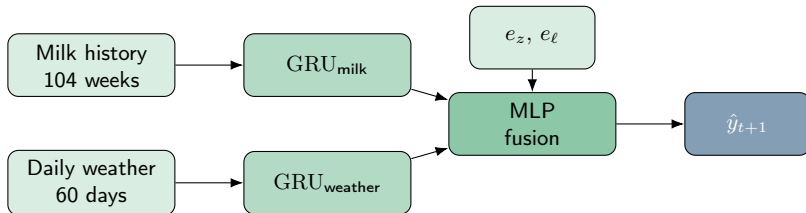


$$\hat{f} : \left(\mathbb{R}^p \times \mathbb{R}^{p' \times d} \times \mathcal{Z} \times \mathcal{L} \right) \rightarrow \mathbb{R}^o$$

$$\{\hat{y}_{t,z,\ell}\}_{t=T}^{T+o-1} = \hat{f}(y_{T-o,z,\ell}, \dots, y_{T-1,z,\ell}, X_{T'-o',z}, \dots, X_{T'-1,z}, z, \ell)$$

Two forecast horizons considered: $o = 12$ and $o = 26$ weeks

Proposed Architecture



Each signal keeps its native resolution ; only the milk branch feeds back its own forecasts ; weather stays externally fed throughout the horizon

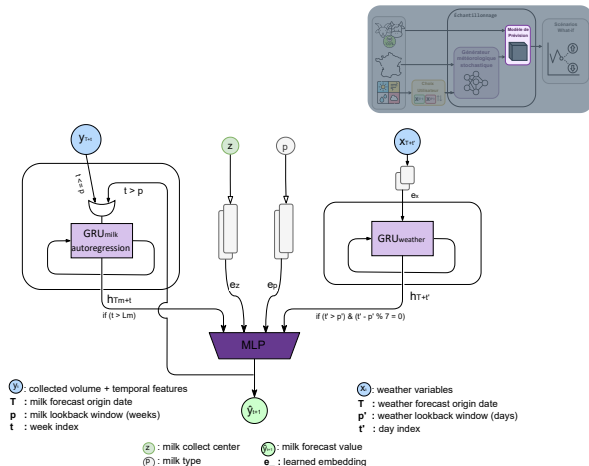
Architecture Formalization

$$h_t^m = \text{GRU}_{\text{milk}}(y_t, h_{t-1}^m)$$

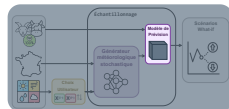
$$h_{t'}^w = \text{GRU}_{\text{weather}}(x_{t'}, h_{t'-1}^w)$$

$$\hat{y}_{t+1} = \text{MLP}(h_t^m, h_{t'}^w, e_z, e_\ell)$$

- Weather hidden state read every 7 days
- Only the milk branch feeds back its own predictions



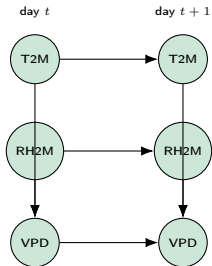
Results : Test Year 2025



Model	MAPE 12 w. (%)	MAPE 26 w. (%)
SARIMA	21.85	23.77
GRU (milk only)	6.60	9.97
Human expert	5.24	7.94
Naively-aligned GRU	4.81	8.45
Dual-branch GRU (ours)	4.50	7.89

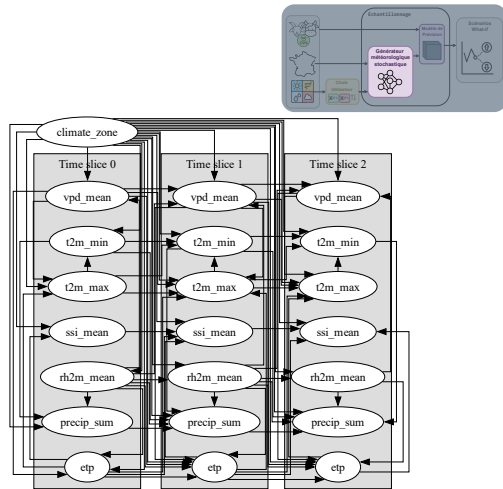
Best performance among all compared methods, including the human expert

Learned Dynamic Bayesian Network



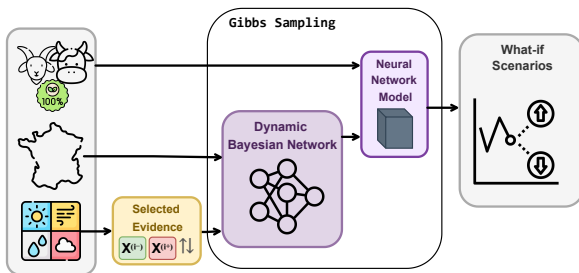
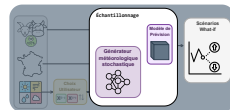
$$P(X_{1:N} | c) = P(X_1 | c) \prod_{t=2}^N P(X_t | \text{Pa}(X_t), c)$$

Structure and conditional distributions learned for each climate zone c — 7 variables, $N = 3$



What-if Mechanism Pipeline

Once the models are trained, the goal is to generate plausible what-if scenarios to query the forecasting model.



- 1 Perturbation: evidence fixed by the user
- 2 Propagated through the DBN and sampled using Gibbs sampling (MCMC) → weather scenarios
- 3 Each scenario is fed into the forecasting model
- 4 Results: scenarios selected based on plausibility and impact

Case Study : Heat Wave Scenario



Contributions & Future Work

- 1 Parallel-branch architecture preserving native resolution per signal
- 2 Dynamic Bayesian Network learning joint weather dynamics per climate zone
- 3 Plausibility-constrained what-if generation via Gibbs sampling

Complementary contributions : forecast accuracy conditions scenario reliability

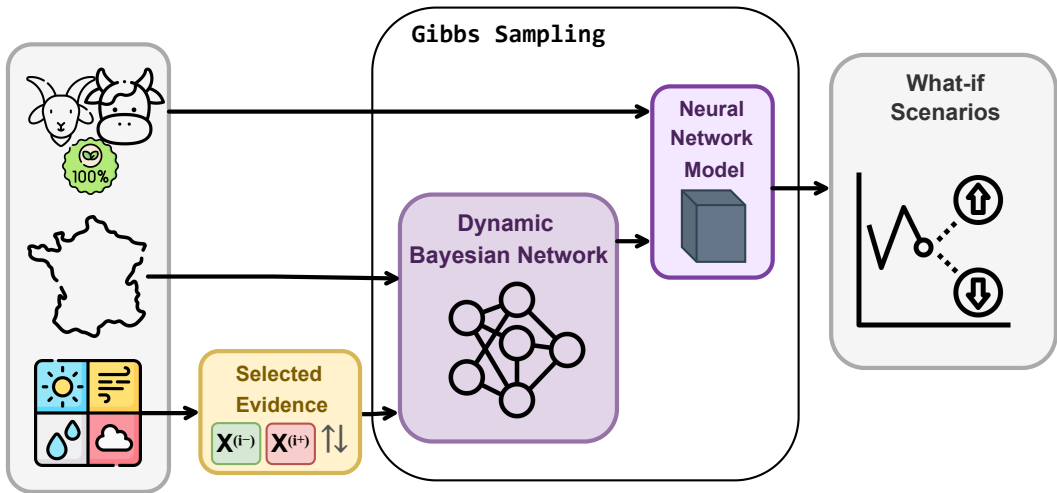
Limitations : Gibbs sampling cost, no retrospective mode, generalizability untested

Future work : amortized sampling, additional agronomic signals, cross-domain generalization

References

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Thank you



Notation

Sets and indices

- $z \in \mathcal{Z}$: collection zone, $\mathcal{Z} = z_1, z_2, \dots$
- $\ell \in \mathcal{L}$: milk type,
 $\mathcal{L} = \text{organic/conventional cow, goat, } \dots$
- t : time index, week
- t' : time index, day

Time series

- $y_{t,z,\ell} \in \mathbb{R}^+ \geq 0$: milk collection
- $X * t', z \in \mathbb{R}^d$: daily weather (d : number of weather variables)

Time windows

- T / T' : common forecast origin, in weeks / days
- p / p' : history length (*lookback*), in weeks / days
- o / o' : forecast horizons, in weeks / days, with $o' = 7o$

